for the PSK so a server must maintain a mapping between PSKs and the associated client or rely on another method for authenticating the client's identity.

> **WARNING** Although TLS allows the PSK to be any length, you should only use a PSK that is cryptographically strong, such as a 128-bit value from a secure random number generator. PSK cipher suites are not suitable for use with passwords because an attacker can perform an offline dictionary or brute-force attack after seeing one PSK handshake.

12.2.1 Implementing a PSK server

Listing 12.8 shows how to load a PSK from a keystore. For this example, you can load the existing HMAC key that you created in chapter 6 but it is good practice to use distinct keys for different uses within an application even if they happen to use the same algorithm. A PSK is just a random array of bytes, so you can call the `getEncoded()` method to get the raw bytes from the `Key` object. Create a new file named PskServer.java under src/main/java/com/manning/apisecurityinaction and copy in the contents of the listing. You'll flesh out the rest of the server in a moment.

Listing 12.8 Loading a PSK

```
package com.manning.apisecurityinaction;

import static java.nio.charset.StandardCharsets.UTF_8;
import java.io.FileInputStream;
```